S.No: 6	Exp. Name: ***Handling Exceptions***	Date:2026-10-17

Aim:

Create a Java program that takes two integer inputs a and b from the user and calculates a/b. Implement exception handling to catch potential arithmetic and input mismatch errors, providing appropriate error messages for each case. Your task is to write a Java Program to Handle **Arithmetic Exceptions** and **InputMisMatchExceptions**.

Input Format:

- The input consists of two numbers each in a new line (integer).

Output Format:

- The output represents the result of the division of the two numbers or any error that occurred while dividing.

Constraints:

- $0 \leq a \leq 10^2$
- $0 \leq b \leq 10^2$

Note:

- The print statements for the output are already present in the editor. Your task is to implement the logic in the specified sections of the code.
- Refer to the visible test cases to strictly match the input/output layout.

Source Code:

ExceptionHandlingExample.java

```
import java.util.InputMismatchException;
import java.util.Scanner;

public class ExceptionHandlingExample {
    public static void main(String[] args) {
        Scanner input = new Scanner(System.in);
        int a, b;

        try {
            System.out.print("Enter the value of 'a': ");
            a = input.nextInt();

            System.out.print("Enter the value of 'b': ");
            b = input.nextInt();

            int result = a / b;
            System.out.println("Result of a/b: " + result);// Complete the code...


        } catch (InputMismatchException e) {
            // Complete the code...
           System.out.println("Input Mismatch Exception: Please enter valid integer v
alues.");
```

```
        } catch (ArithmeticException e) {
             System.out.println("Arithmetic Exception: / by zero");
            //System.out.println("Input Mismatch Exception: Please enter valid intege
r values.");
        }

        input.close();
    }

    public static int divide(int a, int b) {
        return a / b;
    }
}
```


Execution Results - All test cases have succeeded!

Test Case - 1
User Output
Enter the value of 'a':
1
Enter the value of 'b':
0
Arithmetic Exception: / by zero

Test Case - 2
User Output
Enter the value of 'a':
s
Input Mismatch Exception: Please enter valid integer values.

Test Case - 3
User Output
Enter the value of 'a':
50
Enter the value of 'b':
4
Result of a/b: 12

Test Case - 4
User Output
Enter the value of 'a':
3.5
Input Mismatch Exception: Please enter valid integer values.